

Module 3: Three-Dimensional Planar Transmon Geometry and FEM

Hard-coded MATLAB geometry for quasiparticle-poisoning mitigation studies

Physics note for FEM_BallisticDiffusion_PINN

Purpose. This note defines the first physical device geometry for the project: a planar transmon-like superconducting qubit structure. The goal is not to simulate quantum gate dynamics. The goal is to create a geometry-resolved thermal and quasiparticle transport model in which trap material, trap placement, pad shape, substrate material, and heat-sink geometry can be changed systematically.

Contents

Symbols and acronyms used in this document	2
1 Why a planar transmon is the right first geometry	2
2 Baseline dimensions	3
3 Important multiscale modeling decision	3
4 Geometry as a set of cuboids	4
5 Scalar FEM problem for heat transport	4
6 Scalar FEM problem for quasiparticles	5
7 Junction poisoning metric	5
8 What should be swept first	5
9 Limitations of the first geometry model	6
10 References	6

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
Transmon	Transmission-line shunted plasma oscillation qubit	Superconducting qubit design consisting of a Josephson junction shunted by a large capacitance.
CPB	Cooper-pair box	Charge qubit ancestor of the transmon.
E_J	Josephson energy	Energy scale associated with tunneling of Cooper pairs across the Josephson junction.
E_C	Charging energy	Energy cost for changing island charge by one Cooper pair.
C_Σ	Total capacitance	Effective shunt capacitance of the qubit island.
Ω	Computational domain	Full 3D body used by the FEM solver.
Ω_{sub}	Substrate region	Sapphire or silicon substrate volume.
Ω_{sc}	Superconducting metal region	Aluminum, tantalum, or niobium pad/lead region.
Ω_{trap}	Trap region	Normal-metal region that removes quasiparticles.
Ω_{JJ}	Junction-sensitive region	Effective small region near the Josephson junction.
h_e	Element size	Characteristic local finite-element length scale.
N_i	Shape function	FEM interpolation basis function associated with node i .
M	Mass matrix	FEM matrix from time-derivative terms.
K	Stiffness matrix	FEM matrix from diffusion terms.
A	Reaction/trap matrix	FEM matrix from local sink or reaction terms.
J_{JJ}	Junction exposure metric	Time-integrated quasiparticle density in Ω_{JJ} .

1 Why a planar transmon is the right first geometry

A planar transmon is a good first geometry because it contains the important physical features without requiring an impossible first mesh. It has superconducting pads, a dielectric substrate, a

small Josephson junction region, and natural locations for quasiparticle traps and heat sinks. It is also geometrically simple enough to hard-code in MATLAB using cuboids.

The transmon design logic is based on making the Josephson energy much larger than the charging energy:

$$\frac{E_J}{E_C} \gg 1. \quad (1.1)$$

The large shunt capacitance reduces sensitivity to offset charge noise. For this project, the most important geometric consequence is that the qubit contains large superconducting capacitor pads connected by a small junction region. That separation lets us ask a meaningful transport question: how does a nonequilibrium quasiparticle population generated in one place reach, avoid, or get removed before reaching the junction?

2 Baseline dimensions

The baseline device is defined by a substrate and several metal cuboids. The first dimensions are intentionally approximate and sweepable.

Region	Material	Dimensions	Modeling role
Substrate	sapphire or high-resistivity silicon	3 mm $\times$ 3 mm $\times$ 0.5 mm	Main 3D body.
Left capacitor pad	Al baseline	600 μm $\times$ 300 μm $\times$ 100 nm	Left superconducting island.
Right capacitor pad	Al baseline	600 μm $\times$ 300 μm $\times$ 100 nm	Right superconducting island.
Pad gap	exposed substrate	60 μm baseline	Capacitive gap.
Junction leads	Al	8 μm wide, 40 μm long	Connect pads to junction.
Effective JJ region	Al/AlOx/Al reduced patch	1 μm $\times$ 1 μm $\times$ 100 nm effective	Sensitive region for metric.
Normal trap	Cu/Au/Pd	100 μm $\times$ 10 μm $\times$ 80 nm	QP sink.
Backside heat sink	Cu/Au	300 μm $\times$ 300 μm $\times$ 1 μm effective	Optional thermal sink.

3 Important multiscale modeling decision

The Josephson junction contains an oxide barrier that can be only a few nanometers thick. The substrate is hundreds of micrometers thick and millimeters wide. If the mesh tried to resolve the oxide thickness directly across the whole structure, the element count would become unreasonable.

Therefore the first implementation uses an effective region model:

$$\Omega_{JJ} = \text{small effective sensitive volume near the junction}, \quad (3.1)$$

not a direct oxide-barrier mesh. Quasiparticle poisoning is measured by integrating density in this effective region. The junction can also be assigned special source, sink, or loss parameters without meshing the barrier explicitly.

4 Geometry as a set of cuboids

Each physical region is represented as an axis-aligned cuboid:

$$\mathcal{B}_r = \{(x, y, z) : x_{r,1} \leq x \leq x_{r,2}, y_{r,1} \leq y \leq y_{r,2}, z_{r,1} \leq z \leq z_{r,2}\}. \quad (4.1)$$

The MATLAB geometry builder stores these cuboids in a structure array. Material tagging is then done by testing the center of each finite element against each cuboid.

This is not as geometrically flexible as CAD, but it is exactly right for the first project stage because it is transparent, version-controllable, and easy to sweep.

5 Scalar FEM problem for heat transport

For a temperature-like unknown $T(\mathbf{r}, t)$, the baseline diffusion equation is

$$C(\mathbf{r}) \frac{\partial T}{\partial t} = \nabla \cdot (k(\mathbf{r}) \nabla T) + Q(\mathbf{r}, t). \quad (5.1)$$

Multiplying by a test function w and integrating over the domain gives

$$\int_{\Omega} w C \frac{\partial T}{\partial t} dV + \int_{\Omega} \nabla w \cdot k \nabla T dV = \int_{\Omega} w Q dV + \int_{\partial\Omega} w k \nabla T \cdot \mathbf{n} dA. \quad (5.2)$$

With finite-element expansion

$$T(\mathbf{r}, t) \approx \sum_{j=1}^N T_j(t) N_j(\mathbf{r}), \quad (5.3)$$

the semi-discrete system is

$$M \frac{d\mathbf{T}}{dt} + K\mathbf{T} = \mathbf{f}(t) + \mathbf{g}(t), \quad (5.4)$$

where

$$M_{ij} = \int_{\Omega} C N_i N_j dV, \quad K_{ij} = \int_{\Omega} k \nabla N_i \cdot \nabla N_j dV. \quad (5.5)$$

6 Scalar FEM problem for quasiparticles

For quasiparticle density, use the reduced diffusion-reaction-trapping equation

$$\frac{\partial n_{\text{qp}}}{\partial t} = \nabla \cdot (D_{\text{qp}} \nabla n_{\text{qp}}) - \Gamma(\mathbf{r}) n_{\text{qp}} + S_{\text{qp}}(\mathbf{r}, t) \quad (6.1)$$

for the first linearized implementation. Here Γ may include a linearized recombination term and a trap term. The weak form gives

$$M_n \frac{d\mathbf{n}}{dt} + K_n \mathbf{n} + A_n \mathbf{n} = \mathbf{s}(t), \quad (6.2)$$

with

$$(M_n)_{ij} = \int_{\Omega} N_i N_j dV, \quad (6.3)$$

$$(K_n)_{ij} = \int_{\Omega} D_{\text{qp}} \nabla N_i \cdot \nabla N_j dV, \quad (6.4)$$

$$(A_n)_{ij} = \int_{\Omega} \Gamma(\mathbf{r}) N_i N_j dV. \quad (6.5)$$

7 Junction poisoning metric

The physically relevant metric is not the maximum density somewhere on the chip. It is the density near the junction integrated over time:

$$J_{\text{JJ}} = \int_0^{t_f} \int_{\Omega_{\text{JJ}}} n_{\text{qp}}(\mathbf{r}, t) dV dt. \quad (7.1)$$

In FEM form this becomes approximately

$$J_{\text{JJ}} \approx \sum_{m=0}^{N_t} \Delta t_m \sum_{e \in \mathcal{E}_{\text{JJ}}} V_e \bar{n}_e^{(m)}, \quad (7.2)$$

where $\mathcal{E}_{\text{JJ}}$ is the set of elements whose centers lie in the junction-sensitive region, V_e is element volume, and $\bar{n}_e^{(m)}$ is the element-average density at time step m .

8 What should be swept first

The first geometry sweeps should be simple and physically interpretable:

1. no trap versus one trap,
2. trap distance from the junction,

3. trap length and width,
4. trap material through Γ_{trap} and thermal conductivity,
5. substrate material,
6. backside heat-sink location and size.

9 Limitations of the first geometry model

This model is a transport model, not a full superconducting quantum circuit simulation. It does not yet solve the electromagnetic mode structure, the capacitance matrix, the participation ratios, or the Josephson nonlinear quantum Hamiltonian. Those can be added later. The first target is narrower and more realistic: reduce quasiparticle exposure near the junction in a geometry-resolved 3D transport model.

10 References

- [1] J. Koch et al., “Charge-insensitive qubit design derived from the Cooper pair box,” *Physical Review A* **76**, 042319 (2007).
- [2] M. Tinkham, *Introduction to Superconductivity*, 2nd ed., Dover (2004).
- [3] O. V. Lounasmaa, *Experimental Principles and Methods Below 1 K*, Academic Press (1974).